MING'S PERSONAL STUDY NOTES

STATS 361: Causal Inference

Lectures 1–10

English translation of the handwritten notebook

How to read this edition

The translation follows the handwritten notes, including explanations, intuitions, questions, derivations, and marginal additions. Main pages and their supplements are joined by topic and by the arrows in the scans, rather than by a fixed odd/even page rule.

Scan references use one-based PDF page numbers. Author annotations preserve remarks visible in the notebook; they do not claim independent verification that a remark is absent from the textbook. Translator's notes identify uncertain handwriting or apparent source notation issues. They are separate from the author's notes. This is a translation, not a replacement textbook or a mathematical errata edition.

Source: `Lecture1-10.pdf`, 58 scanned pages. The accompanying source-order audit records page coverage and unresolved readings.

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Lecture 1: Randomized Controlled Trials

Scan reference: Lecture1-10.pdf, pp. 2, 1, 3; reading order: main text on p. 2, its supplement on p. 1, then p. 3.

Regression in an RCT analysis of the average treatment effect reduces variance; it does not play a role in identifying the ATE.

$$\Delta_i = Y_i(1) - Y_i(0), \qquad \tau = E[Y_i(1) - Y_i(0)].$$

Estimate and construct inference for τ using the difference-in-means estimator:

$$\widehat{\tau}_{\mathrm{DM}} = \frac{1}{n_1} \sum_{W_i=1} Y_i - \frac{1}{n_0} \sum_{W_i=0} Y_i, \qquad n_w = |\{i : W_i = w\}|.$$

The note marks this estimator as consistent and unbiased.

Supplement: assumptions.

$$Y_i = Y_i(W_i), \qquad W_i \perp \{Y_i(0), Y_i(1)\}.$$

The first is labeled SUTVA; the second is random treatment assignment. SUTVA means the stable unit treatment value assumption: (1) unit stability, with experimental units independent of one another; (2) a single treatment level/version. These are the author's short descriptions.

Linear outcome model and variance reduction

$$Y_i(w) = c_{(w)} + X_i \beta_{(w)} + \varepsilon_i(w), \quad E[\varepsilon_i(w) \mid X_i] = 0, \quad \mathrm{Var}[\varepsilon_i(w) \mid X_i] = b^2.$$

The coefficients are fitted separately for the two treatment states. Write $A = \mathrm{Var}[X]$ and $\|v\|_A^2 = v'Av$. The variance in the notes is

$$V_{\mathrm{DM}} = 4b^2 + \|\beta_{(0)} + \beta_{(1)}\|_A^2 + \|\beta_{(0)} - \beta_{(1)}\|_A^2.$$

Also,

$$\tau = c_{(1)} - c_{(0)} + E[X](\beta_{(1)} - \beta_{(0)}), \qquad \widehat{\tau}_{\mathrm{OLS}} = \widehat{c}_{(1)} - \widehat{c}_{(0)} + \overline{X}(\widehat{\beta}_{(1)} - \widehat{\beta}_{(0)}).$$

The error decomposition written in the notebook is

$$\begin{aligned} \widehat{\tau}_{\mathrm{OLS}} - \tau = (\widehat{c}_{(1)} - c_{(1)}) - (\widehat{c}_{(0)} - c_{(0)}) + \overline{X}(\beta_{(1)} - \beta_{(0)}) \\ + \overline{X}(\widehat{\beta}_{(1)} - \widehat{\beta}_{(0)} - \beta_{(1)} + \beta_{(0)}). \end{aligned}$$

The first two terms are annotated $N(0, b^2/n_1)$ and $N(0, b^2/n_0)$; the third is annotated $N(0, \|\beta_{(1)} - \beta_{(0)}\|_A^2/n)$; the last is $O_p(1/n)$. The margin says that uncertainty in the estimated slopes tends to zero. Thus

$$\sqrt{n}(\widehat{\tau}_{\mathrm{OLS}} - \tau) \Rightarrow N(0, V_{\mathrm{OLS}}), \qquad V_{\mathrm{OLS}} = 4b^2 + \|\beta_{(0)} - \beta_{(1)}\|_A^2,$$

$$V_{\mathrm{DM}} = V_{\mathrm{OLS}} + \|\beta_{(0)} + \beta_{(1)}\|_A^2.$$

Translator's note: the displayed error decomposition and intercept statements use centered covariates, although this centering is not explicitly restated on the scanned page. The factors of four correspond to the balanced-allocation calculation in the notes.

OLS under a general outcome regression

The next page says that OLS is better than difference in means in this model. Suppose

$$Y_i(w) = \mu_{(w)}(X_i) + \varepsilon_i(w), \quad E[\varepsilon_i(w) \mid X_i] = 0, \quad \mathrm{Var}[\varepsilon_i(w) \mid X_i] = b^2.$$

For difference in means,

$$\sqrt{n}(\widehat{\tau}_{\mathrm{DM}} - \tau) \Rightarrow N(0, V_{\mathrm{DM}}), \quad V_{\mathrm{DM}} = 4b^2 + 2\,\mathrm{Var}[\mu_{(0)}(X_i)] + 2\,\mathrm{Var}[\mu_{(1)}(X_i)].$$

For OLS, the Huber–White calculation starts from the population projection

$$(c^*_{(w)}, \beta^*_{(w)}) = arg\min_{c,\beta} E[(Y_i(w) - X_i\beta - c)^2].$$

The notebook records

$$\sqrt{n_w}\left\{\begin{pmatrix}\widehat{c}_{(w)}\\ \widehat{\beta}_{(w)}\end{pmatrix} - \begin{pmatrix}c^*_{(w)}\\ \beta^*_{(w)}\end{pmatrix}\right\} \Rightarrow N\left(0, \begin{pmatrix}\mathrm{MSE}^*_{(w)} & 0\\ 0 & \cdots\end{pmatrix}\right),$$

with $c^*_{(w)} = E[Y_i(w)]$ and

$$\mathrm{MSE}^*_{(w)} = E[(Y_i(w) - X_i\beta^*_{(w)} - c^*_{(w)})^2].$$

It concludes

$$\sqrt{n}(\widehat{\tau}_{\mathrm{OLS}} - \tau) \Rightarrow N(0, V_{\mathrm{OLS}}), \qquad V_{\mathrm{OLS}} = V_{\mathrm{DM}} - \|\beta^*_{(0)} + \beta^*_{(1)}\|^2_A.$$

The red annotation says "$V_{\mathrm{OLS}} < V_{\mathrm{DM}}$ always holds." *Translator's note: the displayed squared-norm identity gives a weak inequality, with equality when the norm is zero. The blanket strict inequality and the simplified covariance matrix above are preserved as statements in the original notes.*

Personal observation. OLS can be viewed as

$$\widehat{\tau}_{\mathrm{OLS}} = \frac{1}{n}\sum_{i=1}^{n}\{(\widehat{c}_{(1)} + X_i\widehat{\beta}_{(1)}) - (\widehat{c}_{(0)} + X_i\widehat{\beta}_{(0)})\} = \frac{1}{n}\sum_i[\widehat{\mu}_{(1)}(X_i) - \widehat{\mu}_{(0)}(X_i)].$$

Here $\widehat{\mu}_{(w)}(x)$ is the fitted OLS value at x. It could be replaced with another method.

Lecture 2: Unconfoundedness and the Propensity Score

Scan reference: Lecture1-10.pdf, pp. 5, 4, 7, 6; p. 4 continues the bottom of p. 5, and p. 6 supplements p. 7.

Unconfoundedness: the aim is to remove the influence of confounding factors. Propensity score: the probability of treatment given the covariates.

Aggregating difference-in-means estimators

The red margin describes this as simply weighting by probability. With discrete, numerically valued X_i, assume

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i = x, \qquad x \in \mathcal{X}.$$

Here x is the grouping variable. Define the group-wise ATE

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x].$$

Then

$$\begin{aligned}
\widehat{\tau}_{\text{AGG}} &= \sum_{x \in \mathcal{X}} \frac{n_x}{n} \widehat{\tau}(x), \\
\widehat{\tau}(x) &= \frac{1}{n_{x1}} \sum_{X_i = x, W_i = 1} Y_i - \frac{1}{n_{x0}} \sum_{X_i = x, W_i = 0} Y_i, \\
n_x &= |\{i : X_i = x\}|, \qquad n_{xw} = |\{i : X_i = x, W_i = w\}|.
\end{aligned}$$

Let $e(x) = P(W_i = 1 \mid X_i = x)$. The within-group limit is

$$\begin{aligned}
\sqrt{n_x}(\widehat{\tau}(x) - \tau(x)) &\Rightarrow N\left(0, \frac{\text{Var}[Y_i(0) \mid X_i = x]}{1 - e(x)} + \frac{\text{Var}[Y_i(1) \mid X_i = x]}{e(x)}\right) \\
&= N\left(0, \frac{b^2(x)}{e(x)(1 - e(x))}\right),
\end{aligned}$$

where the second equality uses the common conditional variance written in the notes. Set $\pi(x) = P(X_i = x)$ and $\widehat{\pi}(x) = n_x/n$. Then

$$\begin{aligned}
\widehat{\tau}_{\text{AGG}} = &\sum_x \pi(x)\tau(x) + \sum_x \pi(x)(\widehat{\tau}(x) - \tau(x)) \\
&+ \sum_x (\widehat{\pi}(x) - \pi(x))\tau(x) + \sum_x (\widehat{\pi}(x) - \pi(x))(\widehat{\tau}(x) - \tau(x)).
\end{aligned}$$

The first term is τ. The second is annotated as asymptotically normal, with variance

$$\sum_x \pi^2(x) \, \text{Var}[\widehat{\tau}(x)].$$

The third has variance $n^{-1} \, \text{Var}[\tau(X_i)]$; the last is $O_p(1/n)$. Following the arrow onto the left page,

$$\sqrt{n}(\widehat{\tau}_{\text{AGG}} - \tau) \Rightarrow N(0, V_{\text{AGG}}),$$

$$\begin{aligned}
V_{\text{AGG}} &= \text{Var}[\tau(X_i)] + \sum_{x \in \mathcal{X}} \pi^2(x) \frac{1}{\pi(x)} \frac{b^2(x)}{e(x)(1 - e(x))} \\
&= \text{Var}[\tau(X_i)] + E\left[\frac{b^2(X_i)}{e(X_i)(1 - e(X_i))}\right].
\end{aligned}$$

Continuous covariates and propensity scores

Given X_i, W_i cannot "peek" at Y_i: this is the author's intuition for unconfoundedness. If

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i, \quad e(x) = P(W_i = 1 \mid X_i = x),$$

then

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid e(X_i).$$

Supplementary sketches and calculation. For discrete X, the sketch shows two outcome points for each x_1, x_2, x_3, joined by a downward arrow. The upper values are 10, 8, 7, the lower values 5, 4, 5, and the gaps are 5, 4, 2. For continuous X, the two sets of points become upper and lower

curves; comparisons are made over $x \in \mathcal{X}$ and related to $e(x)$. The proof written beneath the sketches is

$$\begin{aligned} & P(W_i = w \mid \{Y_i(0), Y_i(1)\}, e(X_i)) \\ &= \int_{\mathcal{X}} P(W_i = w \mid \{Y_i(0), Y_i(1)\}, X_i = x) P(X_i = x \mid e(X_i))\, dx \\ &= \int_{\mathcal{X}} P(W_i = w \mid X_i = x) P(X_i = x \mid e(X_i))\, dx \\ &= e(X_i)\mathbf{1}_{\{w=1\}} + (1 - e(X_i))\mathbf{1}_{\{w=0\}}. \end{aligned}$$

The margin labels this "Lebesgue integration." *Translator's note: this reproduces the notebook's informal integral notation; the first conditional mixing distribution omits conditioning on the potential outcomes. The final expression is the intended balancing-score result.*

Propensity stratification

1. Sort the observations so that $\widehat{e}(X_{i_1}) \le \widehat{e}(X_{i_2}) \le \cdots \le \widehat{e}(X_{i_n})$.
2. Split them into J strata and estimate a difference in means within each stratum. If I_j contains ranks $\lfloor (j-1)n/J \rfloor + 1$ through $\lfloor jn/J \rfloor$, write the abbreviated formula on the page as

   $$\widehat{\tau}_j = \frac{\sum_{i \in I_j} W_i Y_i}{\sum_{i \in I_j} W_i} - \frac{\sum_{i \in I_j} (1 - W_i) Y_i}{\sum_{i \in I_j} (1 - W_i)}.$$

   The blue comment says that stratification makes the control and treatment groups more similar.
3. Average the stratum estimates: $\widehat{\tau}_{\text{STRAT}} = J^{-1} \sum_{j=1}^{J} \widehat{\tau}_j$. The notebook calls this consistent.

Inverse-propensity weighting

Scan reference: Lecture1-10.pdf, pp. 9, 8, 10; p. 8 supplies the model used in the variance calculation on p. 9.

$$\widehat{\tau}_{\text{IPW}} = \frac{1}{n} \sum_{i=1}^{n} \left\{ \frac{W_i Y_i}{\widehat{e}(X_i)} - \frac{(1 - W_i) Y_i}{1 - \widehat{e}(X_i)} \right\}.$$

Oracle IPW assumes $e(x)$ is perfectly known:

$$\widehat{\tau}^*_{\text{IPW}} = \frac{1}{n} \sum_{i} \left\{ \frac{W_i Y_i}{e(X_i)} - \frac{(1 - W_i) Y_i}{1 - e(X_i)} \right\}.$$

Suppose overlap holds, $\eta \le e(x) \le 1 - \eta$ for all $x \in \mathcal{X}$, and $|Y_i| < M$. If

$$\sup_{x \in \mathcal{X}} |e(x) - \widehat{e}(x)| = O_p(a_n) \longrightarrow 0,$$

the notebook gives

$$|\widehat{\tau}_{\text{IPW}} - \widehat{\tau}^*_{\text{IPW}}| = O_p(a_n M / \eta).$$

Thus consistency of the oracle estimator transfers to estimated-propensity IPW. Moreover,

$$E[\widehat{\tau}^*_{\text{IPW}}] = E[Y_i(1) - Y_i(0)] = \tau.$$

With overlap, oracle IPW converges at the $1/\sqrt{n}$ scale.

Variance calculation: supplementary model. The left page supplies

$$\begin{aligned} Y_i(0) &= c(X_i) - (1 - e(X_i))\tau(X_i) + \varepsilon_i(0), \\ Y_i(1) &= c(X_i) + e(X_i)\tau(X_i) + \varepsilon_i(1), \\ E[\varepsilon_i(w) \mid X_i] &= 0, \qquad \mathrm{Var}[\varepsilon_i(w) \mid X_i = x] = b^2(x). \end{aligned}$$

Then the notes state

$$\begin{aligned} n\,\mathrm{Var}[\widehat{\tau}^*_{\mathrm{IPW}}] = V_{\mathrm{IPW}^*} &= E\left[\frac{c^2(X_i)}{e(X_i)(1-e(X_i))}\right] + \mathrm{Var}[\tau(X_i)] \\ &\quad + E\left[\frac{b^2(X_i)}{e(X_i)(1-e(X_i))}\right], \\ \sqrt{n}(\widehat{\tau}^*_{\mathrm{IPW}} - \tau) &\Rightarrow N(0, V_{\mathrm{IPW}^*}). \end{aligned}$$

Personal comparison. $c(x)$ describes the relationship between X and Y, rather than the relationship between X and ΔY. Compared with aggregation,

$$V_{\mathrm{IPW}^*} = V_{\mathrm{AGG}} + E\left[\frac{c^2(X_i)}{e(X_i)(1-e(X_i))}\right].$$

Unless $c(X_i)$ is zero everywhere, oracle IPW has worse variance than aggregation. Compared with linear modeling: IPW is based on $\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i$, while the linear model relies on linearity. *Translator's note: the two displayed potential-outcome equations and the claimed c^2 variance expression are transcribed as written. The weighting defining $c(x)$ needs checking: the c^2 expression ordinarily uses $c(x) = (1-e(x))\mu_1(x) + e(x)\mu_0(x)$, whereas the scanned equations use the opposite coefficients.*

Lecture 3: Efficient Treatment Effect Estimation via Augmented IPW

Scan reference: Lecture1-10.pdf, pp. 11–12.

Ordinary IPW has poor large-sample properties. For example, with discrete X, it is worse than the aggregate estimator $\widehat{\tau}$.

Two characterizations of the ATE

1. Using $e(x) = P(W_i = 1 \mid X_i = x)$, $\tau = E[\widehat{\tau}^*_{\mathrm{IPW}}]$, where the oracle estimator is as above.
2. Using $\mu_{(w)}(x) = E[Y_i(w) \mid X_i = x]$,

$$\tau(x) = \mu_{(1)}(x) - \mu_{(0)}(x), \quad \tau = E[\mu_{(1)}(X_i) - \mu_{(0)}(X_i)],$$

which motivates

$$\widehat{\tau}_{\mathrm{REG}} = \frac{1}{n}\sum_i [\widehat{\mu}_{(1)}(X_i) - \widehat{\mu}_{(0)}(X_i)].$$

The augmented IPW estimator

$$\begin{aligned} \widehat{\tau}_{\mathrm{AIPW}} = \frac{1}{n}\sum_i \Bigg[&\widehat{\mu}_{(1)}(X_i) - \widehat{\mu}_{(0)}(X_i) + W_i\frac{Y_i - \widehat{\mu}_{(1)}(X_i)}{\widehat{e}(X_i)} \\ &- (1 - W_i)\frac{Y_i - \widehat{\mu}_{(0)}(X_i)}{1 - \widehat{e}(X_i)}\Bigg]. \end{aligned}$$

Personal intuition. First use $\mu(x)$ to estimate the outcome, and then use IPW on the regression residuals. IPW makes the covariate composition of the data "fair."

Double robustness

AIPW is consistent if either the outcome regressions $\widehat{\mu}_{(w)}(x)$ are consistent or $\widehat{e}(x)$ is consistent. When $\widehat{\mu}_{(w)}(x) \approx \mu_{(w)}(x)$, the regression average is consistent and the weighted residual terms have mean approximately zero because

$$E[Y_i - \widehat{\mu}_{(W_i)}(X_i) \mid X_i, W_i] \approx 0.$$

The propensity term is held fixed by this conditioning. When instead $\widehat{e}(x) \approx e(x)$, rearrange:

$$\begin{aligned}\widehat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_i \left[\frac{W_i Y_i}{\widehat{e}(X_i)} - \frac{(1-W_i)Y_i}{1-\widehat{e}(X_i)}\right] \\ + \frac{1}{n}\sum_i \left[\widehat{\mu}_{(1)}(X_i)\left(1 - \frac{W_i}{\widehat{e}(X_i)}\right) - \widehat{\mu}_{(0)}(X_i)\left(1 - \frac{1-W_i}{1-\widehat{e}(X_i)}\right)\right].\end{aligned}$$

The first part is IPW. The second has mean approximately zero since $E[1 - W_i/\widehat{e}(X_i) \mid X_i] \approx 0$; conditioning holds the fitted outcome regression fixed.

Semiparametric efficiency

The notes call AIPW asymptotically optimal among nonparametric estimators. When the nuisance functions are estimated sufficiently accurately,

$$\sqrt{n}(\widehat{\tau}_{\text{AIPW}} - \widehat{\tau}^*_{\text{AIPW}}) \longrightarrow_p 0,$$

where the starred estimator is oracle AIPW. This oracle estimator is just an IID average, so the central limit theorem also applies to the estimated version:

$$\sqrt{n}(\widehat{\tau}_{\text{AIPW}} - \tau) \Rightarrow N(0, V^*),$$

$$V^* = \text{Var}[\tau(X_i)] + E\left[\frac{b_0^2(X_i)}{1-e(X_i)}\right] + E\left[\frac{b_1^2(X_i)}{e(X_i)}\right].$$

The margin describes this as a Cramer–Rao-type constraint: this is the best attainable effect/variance.

AIPW and cross-fitting

Scan reference: Lecture1-10.pdf, pp. 13–14.

Randomly split the data into two halves, I_1 and I_2. The annotation says to split the data and choose the halves at random. Set

$$\widehat{\tau}_{\text{AIPW}} = \frac{|I_1|}{n}\widehat{\tau}^{I_1} + \frac{|I_2|}{n}\widehat{\tau}^{I_2},$$

$$\begin{aligned}\widehat{\tau}^{I_1} = \frac{1}{|I_1|}\sum_{i\in I_1}\left[\widehat{\mu}^{I_2}_{(1)}(X_i) - \widehat{\mu}^{I_2}_{(0)}(X_i) + W_i\frac{Y_i - \widehat{\mu}^{I_2}_{(1)}(X_i)}{\widehat{e}^{I_2}(X_i)}\right. \\ \left. - (1-W_i)\frac{Y_i - \widehat{\mu}^{I_2}_{(0)}(X_i)}{1-\widehat{e}^{I_2}(X_i)}\right].\end{aligned}$$

Asymptotically, we can ignore the specific properties of the chosen machine-learning adjustment method and instead rely on:

1. Overlap: $\eta < e(x) < 1 - \eta$ for all $x \in \mathcal{X}$, with $\eta > 0$.

2. Consistency of the nuisance estimators:
$$\sup_x |\widehat{\mu}^{I_2}_{(w)}(x) - \mu_{(w)}(x)| \longrightarrow_p 0, \quad \sup_x |\widehat{e}^{I_2}(x) - e(x)| \longrightarrow_p 0.$$
The blue sentence says that the machine-learning adjustment methods have large-sample consistency.
3. Risk decay for the outcome and propensity models:
$$E[(\widehat{\mu}^{I_2}_{(w)}(X_i) - \mu_{(w)}(X_i))^2]\, E[(\widehat{e}^{I_2}(X_i) - e(X_i))^2] = o(1/n).$$
If both are $\sqrt{n}$-consistent, the product is of order $1/n^2$. A simple sufficient approach is for both regression adjustments to converge at $o(n^{-1/4})$ in RMSE.

Proof recorded on the continuation page. Write the oracle estimator as $\widehat{\tau}^* = |I_1|\widehat{\tau}^{I_1,*}/n + |I_2|\widehat{\tau}^{I_2,*}/n$. To establish $\sqrt{n}(\widehat{\tau}_{\text{AIPW}} - \widehat{\tau}^*) \to_p 0$, it suffices to show the corresponding result for the treated mean; the other components are analogous. Define
$$\widehat{\tau}^{I_1} = \widehat{\mu}^{I_1}_{(1)} - \widehat{\mu}^{I_1}_{(0)}, \qquad \widehat{\mu}^{I_1}_{(1)} = \frac{1}{|I_1|}\sum_{i\in I_1}\left[\widehat{\mu}^{I_2}_{(1)}(X_i) + W_i\frac{Y_i - \widehat{\mu}^{I_2}_{(1)}(X_i)}{\widehat{e}^{I_2}(X_i)}\right].$$
Subtracting its oracle version gives
$$\begin{aligned}\widehat{\mu}^{I_1}_{(1)} - \widehat{\mu}^{I_1,*}_{(1)} = {}& \frac{1}{|I_1|}\sum_{i\in I_1}(\widehat{\mu}^{I_2}_{(1)}(X_i) - \mu_{(1)}(X_i))\left(1 - \frac{W_i}{e(X_i)}\right)\\ &+ \frac{1}{|I_1|}\sum_{i\in I_1} W_i(Y_i - \mu_{(1)}(X_i))\left(\frac{1}{\widehat{e}^{I_2}(X_i)} - \frac{1}{e(X_i)}\right)\\ &- \frac{1}{|I_1|}\sum_{i\in I_1} W_i(\widehat{\mu}^{I_2}_{(1)}(X_i) - \mu_{(1)}(X_i))\left(\frac{1}{\widehat{e}^{I_2}(X_i)} - \frac{1}{e(X_i)}\right).\end{aligned}$$
The blue annotations bound the first contribution by an expression written as $o_p(1)/n$ and the last by $o_p(1/\sqrt{n})$. *Translator's note: it is unclear whether the first handwritten bound is intended for the term or its variance. The target statement is the $o_p(n^{-1/2})$ remainder stated above.* For K folds, use nuisance estimates trained without the fold $k(i)$ containing observation i:
$$\begin{aligned}\widehat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_i \Bigg[&\widehat{\mu}^{(-k(i))}_{(1)}(X_i) - \widehat{\mu}^{(-k(i))}_{(0)}(X_i) + W_i\frac{Y_i - \widehat{\mu}^{(-k(i))}_{(1)}(X_i)}{\widehat{e}^{(-k(i))}(X_i)}\\ &- (1 - W_i)\frac{Y_i - \widehat{\mu}^{(-k(i))}_{(0)}(X_i)}{1 - \widehat{e}^{(-k(i))}(X_i)}\Bigg].\end{aligned}$$

Confidence intervals

Scan reference: Lecture1-10.pdf, pp. 16, 15; p. 15 supplies definitions used on p. 16.

Supplement: terminology. The notebook labels the likelihood derivative an efficient score:
$$S(\theta) = \frac{d\log L(\theta)}{d\theta},$$
where $L(\theta)$ is the likelihood. The empirical variance of n sample scores is
$$\frac{1}{n-1}\sum_i (S(\theta_i) - \overline{S(\theta)})^2.$$
Efficient variance is the smallest attainable variance for a given problem, with a Cramer–Rao lower-bound interpretation. *Translator's note: the derivative shown is the ordinary likelihood score; calling it the efficient score requires the relevant model/orthogonalization context.*

Variance estimation. Define the oracle summand

$$\psi_i^* = \mu_{(1)}(X_i) - \mu_{(0)}(X_i) + \frac{W_i(Y_i - \mu_{(1)}(X_i))}{e(X_i)} - \frac{(1-W_i)(Y_i - \mu_{(0)}(X_i))}{1-e(X_i)}.$$

Its empirical variance converges to the efficient variance:

$$\frac{1}{n-1}\sum_i (\psi_i^* - \widehat{\tau}^*)^2 \longrightarrow_p V^*.$$

The margin calls $\widehat{\tau}^*$ the non-cross-fitted oracle AIPW estimator. Replacing each nuisance function by the estimate trained outside $k(i)$ gives the cross-fitted summand $\widehat{\psi}_i$ and

$$\widehat{V}_{\text{AIPW}} = \frac{1}{n-1}\sum_i (\widehat{\psi}_i - \widehat{\tau}_{\text{AIPW}})^2 \longrightarrow_p V^*.$$

The confidence interval drawn under a central normal probability region is

$$\widehat{\tau}_{\text{AIPW}} \pm \frac{1}{\sqrt{n}}\Phi^{-1}(1-\alpha/2)\sqrt{\widehat{V}_{\text{AIPW}}}.$$

Translator's note: the handwritten text says "α-level," while the formula and shaded sketch correspond to coverage $1-\alpha$.

Lecture 4: Estimating Treatment Heterogeneity

Scan reference: Lecture1-10.pdf, pp. 17–18.

For observations (X_i, Y_i, W_i) with $X_i \in \mathcal{X}$, we previously pursued the ATE $E[Y_i(1) - Y_i(0)]$. Heterogeneity analysis instead considers the conditional average treatment effect (CATE):

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x].$$

The blue note describes this as the average for a particular category of X_i.

Regularization bias

Under $\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i$,

$$\tau(x) = \mu_{(1)}(x) - \mu_{(0)}(x), \qquad \mu_{(w)}(x) = E[Y_i \mid X_i = x, W_i = w].$$

A direct CATE estimator is $\widehat{\tau}_T(x) = \widehat{\mu}_{(1)}(x) - \widehat{\mu}_{(0)}(x)$.

Personal concern. If there are too few observations for $\widehat{\mu}_{(1)}$, it may be heavily regularized (toward a straight line). If there are many observations for $\widehat{\mu}_{(0)}$, that fit may be accurate; then the difference $\widehat{\tau}_T(x)$ goes wrong because of the first fit.

Semiparametric modeling

$$Y_i(w) = f(X_i) + w\tau(X_i) + \varepsilon_i(w), \qquad P(W_i = 1 \mid X_i) = e(X_i), \qquad \tau(x) = \psi(x)\beta.$$

f is a function of X_i, and $\psi(x)$ is a nonparametric representation of X; β is the treatment-effect coefficient. Under unconfoundedness, the Robinson transformation is

$$Y_i - m(X_i) = (W_i - e(X_i))\psi(X_i)\beta + \varepsilon_i, \quad m(x) = E[Y_i \mid X_i = x] = f(x) + e(x)\tau(x),$$

with $\varepsilon_i = \varepsilon_i(W_i)$. Design a residual-on-residual OLS regression using

$$\widetilde{Y}_i^* = Y_i - m(X_i), \qquad \widetilde{Z}_i^* = (W_i - e(X_i))\psi(X_i).$$

Regress $\widetilde{Y}_i^*$ on $\widetilde{Z}_i^*$ to obtain $\widehat{\beta}_R^*$ (the notebook uses a coefficient symbol resembling ζ_R). These transformations use only the observed data and the nuisance functions; the margin calls this data preprocessing. The notebook gives

$$\sqrt{n}(\widehat{\beta}_R^* - \beta) \Rightarrow N(0, V_R),$$
$$V_R = \mathrm{Var}[\widetilde{Z}_i^*]^{-1}\,\mathrm{Var}[\widetilde{Z}_i^*\widetilde{Y}_i^*]\,\mathrm{Var}[\widetilde{Z}_i^*]^{-1}.$$

Under homoskedasticity, $\mathrm{Var}(\varepsilon_i \mid X_i, W_i) = b^2$ is constant and V_R is the semiparametrically efficient variance for β. Under heteroskedasticity, the notes say that V_R still applies but is not the efficient variance. *Translator's note: the middle sandwich term above is reproduced literally from the scan; a regression sandwich calculation ordinarily uses the regression residual, rather than the unadjusted transformed outcome.*

Feasible cross-fitted version. The preceding regression is idealized because $m(X_i)$ and $e(X_i)$ are unknown. Fit nonparametric regressions $Y \sim X$ and $W \sim X$; omit the fold containing observation i when constructing

$$\widetilde{Y}_i = Y_i - \widehat{m}^{(-k(i))}(X_i), \qquad \widetilde{Z}_i = (W_i - \widehat{e}^{(-k(i))}(X_i))\psi(X_i).$$

Regress $\widetilde{Y}_i$ on $\widetilde{Z}_i$ to obtain $\widehat{\beta}_R$. If both nuisance RMSEs are $o_p(n^{-1/4})$, the notes conclude

$$\sqrt{n}(\widehat{\beta}_R^* - \widehat{\beta}_R) \longrightarrow_p 0.$$

A loss function for treatment heterogeneity

Scan reference: Lecture1-10.pdf, pp. 20, 19; p. 19 supplements the residual regression and the R-loss on p. 20.

The previous estimator assumes $\tau(x) = \psi(x)\beta$. Often the relationship is not linear, so use machine learning. With

$$\mu_{(w)}(x) = E[Y(w) \mid X = x], \quad \varepsilon_i(w) = Y_i(w) - [\mu_{(0)}(X_i) + w\tau(X_i)],$$

unconfoundedness yields $E[\varepsilon_i(w) \mid X_i, W_i] = 0$. Here $\mu_{(0)}$ plays the role of the previous f. Again,

$$Y_i - m(X_i) = (W_i - e(X_i))\tau(X_i) + \varepsilon_i, \quad m(x) = \mu_{(0)}(x) + e(x)\tau(x), \quad \varepsilon_i = \varepsilon_i(W_i).$$

Equivalently,

$$\tau(\cdot) = \arg\min_t E[((Y_i - m(X_i)) - (W_i - e(X_i))t(X_i))^2].$$

If m and e are known, minimizing this gives $\tau(\cdot)$. The oracle penalized estimator is

$$\widehat{\tau}_R^*(\cdot) = \arg\min_t \left\{ \frac{1}{n}\sum_i [(Y_i - m(X_i)) - (W_i - e(X_i))t(X_i)]^2 + \Lambda_n(t) \right\}.$$

Λ_n regularizes the complexity of the treatment-effect function, for example with lasso or ridge regression; the margin also mentions CNNs. Usually m and e are unknown (the notes parenthetically mention RCTs), so cross-fit them:

$$\widehat{L}_n(t) = \frac{1}{n}\sum_i [(Y_i - \widehat{m}^{(-k(i))}(X_i)) - (W_i - \widehat{e}^{(-k(i))}(X_i))t(X_i)]^2,$$
$$\widehat{\tau}_R = \arg\min_t \{\widehat{L}_n(t) + \Lambda_n(t)\}.$$

For lasso, $\widehat{\tau}(x) = x\widehat{\beta}$, with

$$\widehat{\beta} = \arg\min_\beta \left\{ \frac{1}{n}\sum_i [(Y_i - \widehat{m}^{(-k(i))}(X_i)) - (W_i - \widehat{e}^{(-k(i))}(X_i))X_i\beta]^2 + \lambda\|\beta\|_1 \right\}.$$

Personal supplementary observation. With a constant treatment effect, $Y_i(w) = f(X_i) + \tau w + \varepsilon_i$ and $E[\varepsilon_i \mid X_i, W_i] = 0$, the residual estimator is

$$\widehat{\tau}_R = \frac{\sum_i (Y_i - \widehat{m}^{(-k(i))}(X_i))(W_i - \widehat{e}^{(-k(i))}(X_i))}{\sum_i (W_i - \widehat{e}^{(-k(i))}(X_i))^2}. \qquad \text{(4.5 in the notes)}$$

The premise is that the nuisance components converge fast enough. Even if τ is not constant, this estimator converges to a nonnegatively weighted average of $\tau(x)$ (the annotation also writes a treated-minus-control mean comparison). When heterogeneity is small and overlap presents difficulties, use (4.5) as a replacement. The arrow connects this observation to the R-loss.

Lecture 5: Regression Discontinuity Designs

Scan reference: Lecture1-10.pdf, pp. 22, 21; p. 21 explains why the earlier IPW approach does not apply.

Introduce a cutoff using a running variable $Z_i \in \mathbb{R}$:

$$W_i = \mathbf{1}_{\{Z_i \geq c\}}, \qquad c \text{ is the threshold.}$$

As $Z_i \to c$, observations near c have similar characteristics and differ in whether they receive the intervention. Let $\mu_{(w)}(z) = E[Y_i(w) \mid Z_i = z]$. If the two conditional means are continuous, the CATE at $z = c$ is

$$\tau_c = \mu_{(1)}(c) - \mu_{(0)}(c) = \lim_{z \to c^+} E[Y_i \mid Z_i = z] - \lim_{z \to c^-} E[Y_i \mid Z_i = z].$$

Personal supplement: why does ordinary IPW not apply? Earlier we assumed unconfoundedness $\{Y_i(0), Y_i(1)\} \perp W_i \mid Z_i$ and overlap $\eta \leq P(W_i = 1 \mid Z_i) \leq 1 - \eta$. In RDD, unconfoundedness holds, but overlap fails because the treatment probability given $Z_i = z$ is either zero or one. The blue comment says this fails regardless of the nondegenerate choice of η. We must compare observations near c, using similarity rather than overlap. Failure of overlap means τ_c cannot be estimated at the $\sqrt{n}$ rate. Its minimax error converges at a nonparametric rate depending on the smoothness of $\mu_{(w)}(z)$. For example, a uniformly bounded second derivative at c allows an $n^{-2/5}$ estimation-error rate.

Estimation via local linear regression

Choose a small bandwidth $h_n \to 0$ and a symmetric weighting function K. Fit the two conditional means locally. The coefficient on W_i in

$$\arg \min_{a, \tau, \beta_{(0)}, \beta_{(1)}} \sum_i K\left(\frac{|Z_i - c|}{h_n}\right) [Y_i - a - \tau W_i - \beta_{(0)}(Z_i - c)_- - \beta_{(1)}(Z_i - c)_+]^2$$

is $\widehat{\tau}_c$. The intercept a and slopes $\beta_{(w)}$ are nuisance parameters; the handwritten labels identify negative-side and positive-side parts. Common kernels are the window function $K(x) = \mathbf{1}_{\{|x| \leq 1\}}$ and triangular kernel $K(x) = (1 - |x|)_+$.

Estimator properties and smoothness

Scan reference: Lecture1-10.pdf, pp. 23–24.

Quantify continuity by assuming $\mu_{(w)}$ is twice differentiable with a uniformly bounded second derivative:

$$\left| \frac{d^2}{dz^2} \mu_{(w)}(z) \right| \leq B, \qquad z \in \mathbb{R}, \quad w \in \{0, 1\}.$$

If smoothness is weaker, with only Lipschitz continuity (the note says bounded but not differentiable), local averaging may work better because there is no clear linear trend locally. If smoothness is stronger, with bounded derivatives of order $k \geq 3$, higher-order local regression may improve the convergence rate. Taylor-expand at c:

$$\mu_{(w)}(z) = a_{(w)} + \beta_{(w)}(z-c) + \tfrac{1}{2}\rho_{(w)}(z-c), \qquad |\rho_{(w)}(u)| \leq Bu^2.$$

The last term is the remainder, and $\tau_c = a_{(1)} - a_{(0)}$. Fit each side separately:

$$(\widehat{a}_{(1)}, \widehat{\beta}_{(1)}) = arg\min_{a,\beta} \sum_{Z_i \geq c} K\left(\frac{|Z_i - c|}{h_n}\right)[Y_i - a - \beta(Z_i - c)]^2,$$

and analogously on the control side; then $\widehat{\tau}_c = \widehat{a}_{(1)} - \widehat{a}_{(0)}$.

Closed-form weights for the window kernel. On $c \leq Z_i \leq c + h_n$, the scan writes

$$\widehat{a}_{(1)} = \sum_i \gamma_i Y_i, \qquad \gamma_i = \frac{\widehat{E}_{(1)}[(Z_i - c)^2] - \widehat{E}_{(1)}[Z_i - c](Z_i - c)}{\widehat{E}_{(1)}[(Z_i - c)^2] - \widehat{E}_{(1)}[Z_i - c]^2},$$

where $\widehat{E}_{(1)}[Z_i - c]$ is the average of $Z_i - c$ within the bandwidth. It also records

$$\sum_i \gamma_i = 1, \qquad \sum_i \gamma_i (Z_i - c) = 0.$$

Translator's note: the displayed weight in the scan is missing division by the number of observations in the bandwidth. That normalization is required for the subsequent sum-to-one identity. All sums in this paragraph run over the bandwidth. The decomposition on the page is

$$\widehat{a}_{(1)} = a_{(1)} + \underbrace{\sum_i \gamma_i \rho_{(1)}(Z_i - c)}_{\text{curvature bias}} + \underbrace{\sum_i \gamma_i (Y_i - \mu_{(1)}(Z_i))}_{\text{sampling noise}}.$$

The control intercept has an analogous decomposition. The red arrows say that both terms need bounds. Curvature bias is bounded at order Bh_n^2, where B bounds the second derivative and h_n is the bandwidth. Sampling noise has mean zero; if $\text{Var}[Y_i \mid Z_i] \leq b^2$, its variance is bounded by $b^2 \sum_i \gamma_i^2$. If Z_i has a continuous nonzero density f near c,

$$b^2 \sum_i \gamma_i^2 \approx \frac{4b^2}{|\{i : c \leq Z_i \leq c + h_n\}|} \approx \frac{4b^2}{f(c) n h_n}.$$

Thus squared bias is of order h_n^4 and variance is of order $(nh_n)^{-1}$, giving

$$\widehat{\tau}_c = \tau_c + O_p(n^{-2/5}), \qquad h_n \asymp n^{-1/5}.$$

The red note identifies $n^{-2/5}$ as the estimation error and adds that with k-th order smoothness, $h_n \asymp n^{-1/(2k+1)}$. *Translator's note: the remainder decomposition drops the factor $1/2$ used in the preceding Taylor line; this does not change the rate argument.*

Noise in the running variable

Scan reference: Lecture1-10.pdf, p. 25.

The blue margin notes that the running variable may contain noise. Introduce a latent $U_i \sim G$ and two conditions:

1. A noisy running variable: the scan writes $Z_i \mid U_i \sim N(0, \nu^2)$, with $\nu > 0$.
2. Unconfoundedness of the running variable: $\{Y_i(0), Y_i(1)\} \perp Z_i \mid U_i$.

With $Y_i = Y_i(W_i)$ and $\{Y_i(0), Y_i(1)\} \perp W_i \mid U_i$,

$$e(u) = P(W_i = 1 \mid U_i = u) = P(Z_i \geq c \mid U_i = u) = 1 - \Phi\left(\frac{c-u}{\nu}\right),$$

$$\alpha_{(w)}(u) = E[Y_i(w) \mid U_i = u], \qquad \tau(u) = E[Y_i(1) - Y_i(0) \mid U_i = u].$$

The annotation defines the standard normal CDF, $\Phi(x) = (2\pi)^{-1/2} \int_{-\infty}^{x} e^{-t^2/2}\, dt$. *Translator's note: the ensuing propensity and density formulas require $Z_i \mid U_i = u \sim N(u, \nu^2)$, or equivalently $Z_i - U_i \mid U_i \sim N(0, \nu^2)$. The conditional normal line above is preserved as written.* U_i is unobserved, so we cannot directly use its propensity score. Consider a deconvolution-type estimator:

$$\widehat{\tau}_\gamma = \frac{1}{n} \sum_{Z_i \geq c} \gamma_+(Z_i) Y_i - \frac{1}{n} \sum_{Z_i < c} \gamma_-(Z_i) Y_i.$$

Choose the weights to satisfy

$$E[\mathbf{1}_{\{Z_i \geq c\}} \gamma_+(Z_i)] = E[\mathbf{1}_{\{Z_i < c\}} \gamma_-(Z_i)] = 1.$$

Then

$$E[\widehat{\tau}_\gamma] = \underbrace{\int h_+(u)\tau(u)\, dG(u)}_{\text{weighted treatment effect}} + \underbrace{\int (h_+(u) - h_-(u))\alpha_{(0)}(u)\, dG(u)}_{\text{confounding bias}},$$

where the weighting functions are

$$h_+(u) = \int_c^\infty \gamma_+(z)\varphi_\nu(z-u)\, dz, \qquad h_-(u) = \int_{-\infty}^c \gamma_-(z)\varphi_\nu(z-u)\, dz,$$

$$\varphi_\nu(z-u) = \frac{1}{\sqrt{2\pi\nu^2}} \exp\left[-\frac{1}{2}\left(\frac{z-u}{\nu}\right)^2\right].$$

Lecture 6: Finite-Sample Inference in RDDs

Scan reference: Lecture1-10.pdf, pp. 27, 26, 29, 28; p. 26 continues p. 27, and p. 28 explains the weight constraints on p. 29.

Finite-sample RDDs allow a discontinuous/discrete running variable or a more complex cutoff rule. The local-linear regression solution (6.2) can be written as a linear estimator,

$$\widehat{\tau}_c(\gamma) = \sum_{i=1}^n \gamma_i Y_i. \qquad \text{(6.3 in the notes)}$$

The weights depend only on Z_i. If

$$Y_i(w) = \mu_{(w)}(Z_i) + \varepsilon_i(w), \qquad \varepsilon_i(w) \mid Z_i \sim N(0, b^2),$$

then, conditional on $Z_1, \ldots, Z_n$,

$$\widehat{\tau}_c(\gamma) \sim N(\widehat{\tau}_c^*(\gamma), b^2\|\gamma\|_2^2), \qquad \widehat{\tau}_c^*(\gamma) = \sum_i \gamma_i \mu_{(W_i)}(Z_i).$$

Here $W_i = \mathbf{1}_{\{Z_i \geq c\}}$ and $\|\gamma\|_2 = (\gamma_1^2 + \cdots + \gamma_n^2)^{1/2}$ is the Euclidean norm.

Minimax linear estimation

Minimax means minimizing the worst-case MSE. For a linear estimator, the conditional variance is directly accessible as $b^2\|\gamma\|_2^2$ under homoskedasticity. Its bias depends on unknown conditional means:

$$\text{Bias}(\widehat{\tau}_c(\gamma) \mid Z_1, \ldots, Z_n) = \sum_i \gamma_i \mu_{(W_i)}(Z_i) - [\mu_{(1)}(c) - \mu_{(0)}(c)].$$

A smoothness restriction $|\mu''_{(w)}(z)| \leq B$ bounds the bias by

$$I_B(\gamma) = \sup_{|\mu''_{(w)}| \leq B} \left\{ \sum_i \gamma_i \mu_{(W_i)}(Z_i) - [\mu_{(1)}(c) - \mu_{(0)}(c)] \right\}.$$

The blue note says the supremum does not need an absolute value because the second-derivative class is symmetric. Hence

$$|\,\text{Bias}(\widehat{\tau}_c(\gamma) \mid Z_1, \ldots, Z_n)| \leq I_B(\gamma),$$
$$\text{MSE}(\widehat{\tau}_c(\gamma) \mid Z_1, \ldots, Z_n) \leq b^2\|\gamma\|_2^2 + I_B^2(\gamma).$$

The red annotation emphasizes "MSE = variance + squared bias" and that this bound does not depend on the unknown μ. The left-page continuation defines

$$\gamma^B = \arg\min_\gamma \{b^2\|\gamma\|_2^2 + I_B^2(\gamma)\}, \qquad \widehat{\tau}_c(\gamma^B) = \sum_i \gamma_i^B Y_i. \qquad \text{(6.8 in the notes)}$$

The blue comparison says that, at a fixed bias bound $I_B(\gamma)$, this yields smaller MSE than local-linear estimation.

Deriving the minimax linear weights

Decompose

$$\mu_{(w)}(z) = a_{(w)} + \beta_{(w)}(z - c) + \rho_{(w)}(z),$$

where $\rho_{(w)}(c) = \rho'_{(w)}(c) = 0$ and the second derivative is bounded by B. The intercepts and slopes are unrestricted, so $I_B(\gamma)$ can be infinite unless

$$\sum_i \gamma_i W_i = 1, \qquad \sum_i \gamma_i = 0, \qquad \sum_i \gamma_i (Z_i - c)_+ = 0, \qquad \sum_i \gamma_i (Z_i - c)_- = 0.$$

After these constraints, only the remainders need control. Rewrite (6.8) as

$$(\gamma^B, t) = \arg\min_{\gamma, t} \{b^2\|\gamma\|_2^2 + B^2 t^2\},$$

subject to

$$\sum_i \gamma_i W_i \rho_{(1)}(Z_i) + \sum_i \gamma_i (1 - W_i) \rho_{(0)}(Z_i) \leq t$$

for every $\rho_{(w)}$ with $\rho_{(w)}(c) = \rho'_{(w)}(c) = 0$ and $|\rho''_{(w)}(z)| \leq 1$, together with

$$\sum_i \gamma_i W_i = 1, \quad \sum_i \gamma_i = 0, \quad \sum_i \gamma_i W_i (Z_i - c) = 0, \quad \sum_i \gamma_i (1 - W_i)(Z_i - c) = 0. \quad \text{(6.10 in the notes)}$$

The blue instruction is to solve the corresponding dual problem.

Personal supplement: why do the weights sum to zero? The final estimate is a difference. Split its sum into the treated and control parts:

$$\widehat{\tau}_c(\gamma) = \sum_i \gamma_i^{(1)} Y_i + \sum_i \gamma_i^{(2)} Y_i.$$

The treated weights sum to 1 and the control weights sum to -1, so

$$\sum_i \gamma_i = 1 + (-1) = 0, \quad \sum_i \gamma_i^{(1)} = \sum_i \gamma_i W_i = 1, \quad \sum_i \gamma_i^{(2)} = \sum_i \gamma_i (1 - W_i) = -1.$$

The sketch shows a weight profile with a negative spike just to the left of the threshold and a positive spike just to the right; farther away, the weights change sign and then approach zero. The red labels on the original say $Y = 1$ and $Y = 0$, while the equations below them use treatment indicators W.

Inference with linear estimators

Scan reference: Lecture1-10.pdf, pp. 30–31.

Construct a confidence interval for $\widehat{\tau}_c(\gamma^B)$. Let

$$\text{err} = \widehat{\tau}_c - \tau_c, \qquad \text{err} \mid Z_1, \ldots, Z_n \sim N(\text{bias}, b^2 \|\gamma^B\|_2^2),$$

where $|\text{bias}| \leq Bt$ from (6.10). By unimodality of the Gaussian distribution, for $S \sim N(0,1)$,

$$P(|\text{err}| \geq \zeta) \leq P(|Bt + b\|\gamma^B\|_2 S| \geq \zeta).$$

A confidence interval with conditional coverage at least $1 - \alpha$ is

$$I_\alpha = (\widehat{\tau}_c(\gamma^B) - \zeta_\alpha^B, \widehat{\tau}_c(\gamma^B) + \zeta_\alpha^B),$$

$$\zeta_\alpha^B = \inf\{\zeta : P(|Bt + b\|\gamma^B\|_2 S| > \zeta) \leq \alpha\}. \qquad \text{(6.13 in the notes)}$$

The blue comment stresses that conditioning on Z_i makes this work for running variables with any distribution, including nonstandard RDD settings.

Discrete running variables. The restriction $|\mu''_{(w)}(z)| \leq B$ alone need not identify τ_c, but (6.13) remains valid. For continuous Z_i, the interval shrinks at rate $n^{-2/5}$; for discrete Z_i, its length may not tend to zero. When c is outside the support of the running variable, one needs an interpolating function representing $\mu_{(w)}(z)$ and satisfying the second-derivative bound.

Multivariate running variables. Examples include two- or three-dimensional regions and geographic boundaries. A scalar-cutoff RDD cannot be used directly; some reduce the problem to one-dimensional distance, which the note calls less stable. The minimax method can instead be extended directly to multiple dimensions.

Beyond homoskedasticity

When the errors $\varepsilon_i = Y_i - \mu_{(W_i)}(Z_i)$ are no longer homoskedastic Gaussian with variance b^2, the exact normal statement (6.11) fails. Use a central limit theorem to justify

$$\widehat{\tau}_c(\gamma) \mid Z_1, \ldots, Z_n \;\approx\; N\left(\widehat{\tau}_c^*(\gamma), \sum_i \gamma_i^2 \,\text{Var}[Y_i \mid Z_i, W_i]\right). \qquad \text{(6.14 in the notes)}$$

With a valid Gaussian approximation, confidence intervals can still be obtained. Estimate the conditional variance by

$$\widehat{V}_n = \sum_i \gamma_i^2 (Y_i - \widehat{\mu}_{(W_i)}(Z_i))^2,$$

where the fitted mean comes from local-linear regression. The blue note says that if this mean model is wrong, residuals are inflated, making the result too conservative. The minimax criterion (6.8) was for homoskedasticity; under heteroskedasticity, introduce individual variances b_i^2 into (6.10), the problem used to construct the dual.

Lecture 7: Balancing Estimators

Scan reference: Lecture1-10.pdf, pp. 32–33.

The heading links balance to the principle behind estimating $\widehat{e}(x)$. Two aims are noted: improving the propensity and regression functions, and estimating ATEs with high-dimensional covariates. IPW and AIPW use propensity scores; this lecture uses convex optimization to derive how to estimate $e(x)$.

Review of IPW

For IID observations $(X_i, Y_i, W_i) \in \mathcal{X} \times \mathbb{R} \times \{0, 1\}$, the goal is $\tau = \mu_{(1)} - \mu_{(0)}$, where $\mu_{(w)} = E[Y_i(w)]$. Assume

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i, \qquad 0 < \eta \leq e(X_i) \leq 1 - \eta < 1,$$

with $e(x) = P(W_i = 1 \mid X_i = x)$ and positive constant η. Estimate the treated mean using

$$\widehat{\mu}_{\text{IPW}}(1) = \frac{1}{n} \sum_i \frac{W_i Y_i}{\widehat{e}(X_i)}. \qquad \text{(7.3 in the notes)}$$

The notebook labels the following consistency argument "brief, not important" but records three steps:

1. Population balance: under unconfoundedness, $\mu_{(1)} = E[W_i Y_i / e(X_i)]$.
2. Oracle estimator: with the true $e(x)$ it is unbiased; overlap gives finite variance.
3. Feasible approximation: if the estimated propensities also satisfy overlap, the Cauchy–Schwarz inequality gives

$$\begin{aligned} |\widehat{\mu}_{\text{IPW}}(1) - \widehat{\mu}^*_{\text{IPW}}(1)| &\leq \left[\frac{1}{n}\sum_i \left(\frac{1}{\widehat{e}(X_i)} - \frac{1}{e(X_i)}\right)^2\right]^{1/2} \left[\frac{1}{n}\sum_i (W_i Y_i)^2\right]^{1/2} \\ &\leq \frac{1}{\eta^2}\left[\frac{1}{n}\sum_i (\widehat{e}(X_i) - e(X_i))^2\right]^{1/2} \left[\frac{1}{n}\sum_i (W_i Y_i)^2\right]^{1/2}. \end{aligned}$$

The margin recalls $|a \cdot b|^2 \leq \|a\|^2 \|b\|^2$. It adds that the analogous AIPW error is negligible relative to $1/\sqrt{n}$ as $n \to \infty$.

Population versus sample balance

Expand the conditional mean in a predefined basis:

$$\mu_{(w)}(x) = \sum_{j=1}^{\infty} \beta_{j(w)} \psi_j(x).$$

The red note gives polynomial and Fourier bases as examples. Under reasonable regularity conditions,

$$\mu_{(w)} = \sum_{j=1}^{\infty} \beta_{j(w)} E[\psi_j(X_i)].$$

The arrow says to take expectations of both sides; a red question asks whether these assumptions may have a problem. With unconfoundedness, $Y_i = \mu_{(W_i)}(X_i) + \varepsilon_i$ and $E[\varepsilon_i \mid X_i, W_i] = 0$. The population IPW argument is

$$\begin{aligned} E\left[\frac{W_i Y_i}{e(X_i)}\right] &= E\left[\frac{W_i}{e(X_i)} \sum_j \beta_{j(1)} \psi_j(X_i)\right] \\ &= \sum_j \beta_{j(1)} E\left[\frac{W_i \psi_j(X_i)}{e(X_i)}\right] = \sum_j \beta_{j(1)} E[\psi_j(X_i)] = \mu_{(1)}. \end{aligned}$$

The red explanation identifies $E[W_i \mid X_i] = e(X_i)$ as the core of the argument. Thus choose weights so that, in the sample,

$$\frac{1}{n} \sum_i \frac{W_i \psi_j(X_i)}{\widehat{e}(X_i)} \approx \frac{1}{n} \sum_i \psi_j(X_i), \qquad j = 1, 2, \ldots. \tag{7.8 in the notes}$$

A balancing loss for propensity estimation

Scan reference: Lecture1-10.pdf, pp. 35, 34; p. 34 supplies the norm reminders used in the high-dimensional bounds.

Suppose the outcome model is linear, $\mu_{(w)}(x) = x\beta_{(w)}$, and the propensity model is logistic,

$$e(x) = \frac{1}{1 + \exp(-x\theta)}.$$

The blue annotation says that sample balance then involves only the covariates X_i, and the targeted balance concerns means. Equation (7.8) becomes

$$\frac{1}{n} \sum_i (1 + e^{-X_i \widehat{\theta}}) W_i X_i = \frac{1}{n} \sum_i X_i. \tag{7.9 in the notes}$$

This is the KKT condition of the convex problem

$$\widehat{\theta} = \arg\min_{\theta} \frac{1}{n} \sum_i \ell_\theta(X_i, Y_i, W_i), \qquad \ell_\theta(X_i, Y_i, W_i) = W_i e^{-X_i \theta} + (1 - W_i) X_i \theta. \tag{7.10 in the notes}$$

It can therefore be solved by Newton descent. The handwritten text describes a unique solution and says this serves as the loss function. The margin notes that estimating both $\mu_{(0)}$ and $\mu_{(1)}$ requires two propensity models.

ATE estimation with high-dimensional confounders

Suppose $X_i \in \mathbb{R}^p$, possibly $p \gg n$, and $\mu_{(w)}(x) = x\beta_{(w)}$. In high dimensions, it is difficult to estimate $e(x)$ adequately; exact weights are not attainable, so seek approximate balance. By Holder's inequality,

$$\begin{aligned} &\frac{1}{n} \sum_i \frac{W_i \mu_{(1)}(X_i)}{\widehat{e}(X_i)} - \frac{1}{n} \sum_i \mu_{(1)}(X_i) \\ &\quad = \left(\frac{1}{n} \sum_i \frac{W_i X_i}{\widehat{e}(X_i)} - \frac{1}{n} \sum_i X_i \right) \beta_{(1)} \\ &\quad \le \left\| \frac{1}{n} \sum_i \frac{W_i X_i}{\widehat{e}(X_i)} - \frac{1}{n} \sum_i X_i \right\|_\infty \|\beta_{(1)}\|_1. \end{aligned}$$

The first norm is labeled imbalance. The handwritten reminder states Holder's bound $\sum_i |a_i b_i| \leq (\sum_i |a_i|^p)^{1/p} (\sum_i |b_i|^q)^{1/q}$.

Personal interpretation and questions. Good inverse-probability weights make the worst case of imbalance as good as possible. Two problems remain: (1) how to parameterize $\widehat{e}(X_i)$ to obtain sup-norm approximate balance; (2) the bound is meaningful only if $\|\beta_{(1)}\|_1$ is bounded. High-dimensional statistics usually assumes a sparse coefficient vector rather than constraining its 1-norm directly.

Supplement: norm reminders. The left page lists the "0-norm" as $\sum_i X_i^0$, the 1-norm as $\sum_i |X_i|$, the 2-norm as $(\sum_i X_i^2)^{1/2}$, and the infinity norm as $\max_i |X_i|$. It gives the general expression $\|x\|_p = (\sum_i x_i^p)^{1/p}$. *Translator's note: these last formulas retain the handwritten shorthand. The standard 0-norm counts nonzero entries, and the standard general p-norm uses $|x_i|^p$.*

Approximate balancing weights and regression augmentation

Scan reference: Lecture1-10.pdf, pp. 36–37.

For a well-chosen tuning parameter ζ, solve

$$\widehat{\gamma} = \arg\min_{\gamma_i \geq 1} \left\{ \frac{1}{n^2} \|\gamma\|_2^2 + \zeta \left\| \frac{1}{n} \sum_i (\gamma_i W_i - 1) X_i \right\|_\infty^2 \right\}. \qquad \text{(7.12 in the notes)}$$

The red labels identify the L_2 norm, imbalance, and tuning weight. Then use $1/\widehat{e}(X_i) = \widehat{\gamma}_i$ as inverse-propensity weights. Sub-Gaussian concentration inequalities verify, for oracle weights, bounds of the form

$$\frac{1}{n^2} \|1/e(X_i)\|_2^2 \leq \frac{\eta^{-2}}{n}, \qquad \left\| \frac{1}{n} \sum_i (\gamma_i W_i - 1) X_i \right\|_\infty^2 = O_p\left(\frac{\eta^{-2} \log p}{n} \right).$$

The notes say $\widehat{e}(X_i)$ need not estimate $e(X_i)$ well; with a suitable ζ, these inverse weights nonetheless satisfy a bias bound. Next augment the weighted estimator with high-dimensional regression to remove dependence on $\|\beta_{(1)}\|_1$. The augmented balancing estimator is

$$\widehat{\mu}_{\text{AB}}(1) = \frac{1}{n} \sum_i \left[\widehat{\mu}_{(1)}(X_i) + \widehat{\gamma}_i W_i (Y_i - \widehat{\mu}_{(1)}(X_i)) \right], \quad \widehat{\mu}_{(1)}(X_i) = X_i \widehat{\beta}_{(1)}. \qquad \text{(7.14 in the notes)}$$

Its decomposition is

$$\widehat{\mu}_{\text{AB}}(1) = \underbrace{\frac{1}{n} \sum_i X_i \beta_{(1)}}_{\text{mean of } \mu_{(1)}(X_i)} + \underbrace{\frac{1}{n} \sum_i \widehat{\gamma}_i W_i (Y_i - X_i \beta_{(1)})}_{\text{mean-zero noise}} + \left[\frac{1}{n} \sum_i (1 - \widehat{\gamma}_i W_i) X_i \right] (\widehat{\beta}_{(1)} - \beta_{(1)}).$$

The last term is bounded in magnitude by

$$\left\| \frac{1}{n} \sum_i (1 - \widehat{\gamma}_i W_i) X_i \right\|_\infty \|\widehat{\beta}_{(1)} - \beta_{(1)}\|_1.$$

The red intuition is that the weights only need to correct the estimation error $\beta_{(1)} - \widehat{\beta}_{(1)}$, rather than the whole fitted coefficient vector. High-dimensional regression supplies a bound on this error: under sparsity $\|\beta_{(1)}\|_0 \le k$ and a restricted-eigenvalue condition, lasso achieves

$$\|\widehat{\beta}_{(1)} - \beta_{(1)}\|_1 = O_p\left(k\sqrt{\frac{\log p}{n}}\right).$$

Putting the steps together. Suppose the design satisfies the conditions for this bound, $\beta_{(1)}$ is k-sparse, and $k \ll \sqrt{n}/\log p$.

1. Fit lasso on the treated units; the coefficient 1-norm error is $o_p(1/\sqrt{\log p})$.
2. Fit $\widehat{\gamma}$ using (7.12), giving infinity-norm imbalance $O_p(\sqrt{\log(p)/n})$.
3. Use (7.14), and insert these bounds into the decomposition:

$$\widehat{\mu}_{\text{AB}}(1) - \frac{1}{n}\sum_i X_i\beta_{(1)} = \frac{1}{n}\sum_i \widehat{\gamma}_i W_i(Y_i - X_i\beta_{(1)}) + o_p(n^{-1/2}). \qquad \text{(7.17 in the notes)}$$

   The noise term is $O_p(n^{-1/2})$.

A central limit theorem also holds:

$$\frac{\sum_i \widehat{\gamma}_i W_i(Y_i - X_i\beta_{(1)})}{\sqrt{\sum_i \widehat{\gamma}_i^2 W_i^2 (Y_i - X_i\beta_{(1)})^2}} \Rightarrow N(0,1).$$

Personal caveat. This is weaker than expected: it only gives a confidence interval for the sample average $\overline{\mu}_{(1)} = n^{-1}\sum_i X_i\beta_{(1)}$, not for $\mu_{(1)}$ itself. To obtain a result for the population mean, one needs to establish convergence of the estimated weights $\widehat{\gamma}_i$.

Lecture 8: Methods for Panel Data

Scan reference: Lecture1-10.pdf, pp. 39, 38; p. 38 supplies the panel layout and within-unit variance supplement.

Panel (longitudinal) data observe units $i = 1, \dots, n$ at times $t = 1, \dots, T$. The left-page sketch is a matrix with units down the rows and time points $t_1, t_2, \dots, T$ across the columns.

A constant treatment-effect model

Outcomes are $Y_{it} \in \mathbb{R}$ and treatment assignments are $W_{it} \in \{0, 1\}$. The basic model is

$$Y_{it} = Y_{it}(0) + W_{it}\tau, \qquad i = 1, \dots, n, \quad t = 1, \dots, T. \qquad \text{(8.1 in the notes)}$$

This very basic model leaves out treatment heterogeneity and treatment dynamics. The notes explain that the treatment effect is the same for all units and all times; a unit's outcome at t depends only on its treatment at t, not on previous treatment. The red annotations emphasize that τ is unrelated to i, t and that past times do not enter.

The two-way model

$$Y_{it} = \alpha_i + \beta_t + W_{it}\tau + \varepsilon_{it}, \qquad E[\varepsilon \mid \alpha, \beta, W] = 0. \qquad \text{(8.2 in the notes)}$$

α_i and β_t are unit and time shifts. The red comment calls the condition strict exogeneity of treatment, stronger than unconfoundedness. For example, with two periods, let one group have treatment path $(0, 0)$ and another $(0, 1)$. The closed-form difference-in-differences estimate is

$$\widehat{\tau} = \frac{1}{|\{i : W_{i2} = 1\}|}\sum_{W_{i2}=1}(Y_{i2} - Y_{i1}) - \frac{1}{|\{i : W_{i2} = 0\}|}\sum_{W_{i2}=0}(Y_{i2} - Y_{i1}).$$

Comments in the notes. A limitation is that with large n, T and general treatment-assignment distributions, model (8.2) may fail. Analogously, OLS with more than two periods is sometimes also called DID, although the note says there is no closed-form OLS solution. The listed advantage is the intuitive parallel-trends requirement (units differ only in α_i); without parallel trends, it cannot be used. *Translator's note: the phrase about there being no closed-form OLS solution is retained as written; general OLS does have a matrix expression. The handwritten point appears to concern the absence of the simple two-period DID expression.*

Supplement: modeling the noise and estimating variance. Treat the row vectors $\varepsilon_i = (\varepsilon_{i1}, \ldots, \varepsilon_{iT})$ as IID across units, with $\mathrm{Var}[\varepsilon_i] = \Sigma$. Treating every ε_{it} as independent is hard to justify and unreasonable. For $T = 2$, estimate the variance of the differences directly:

$$\widehat{\mathrm{Var}}[\widehat{\tau}] = \frac{\widehat{\mathrm{Var}}[Y_{i2} - Y_{i1} \mid W_{i2} = 1]}{|\{i : W_{i2} = 1\}|} + \frac{\widehat{\mathrm{Var}}[Y_{i2} - Y_{i1} \mid W_{i2} = 0]}{|\{i : W_{i2} = 0\}|}. \qquad \text{(8.4 in the notes)}$$

This remains robust when errors within a row/unit are correlated.

Interactive panel models

Scan reference: Lecture1-10.pdf, page 41, with the matrix intuition on page 40

Consider the interactive panel model

$$Y_{it} = A_{i\cdot} B_{t\cdot}^{\mathsf{T}} + W_{it}\tau + \varepsilon_{it}, \qquad \mathbb{E}[\varepsilon \mid A, B, W] = 0, \tag{8.5}$$

where $A \in \mathbb{R}^{n \times k}$, $B \in \mathbb{R}^{T \times k}$ and k is a rank parameter. Equivalently,

$$Y_{it} = L_{it} + W_{it}\tau + \varepsilon_{it} \qquad \text{for a rank-}k\text{ matrix } L.$$

My left-page matrix intuition. Place the A values down the rows and the B values across the columns. For rank one, their intersection has entry $a_i b_t$; for $k \neq 1$, the entry is a sum $a_{i1}b_{t1} + a_{i2}b_{t2} + \cdots$. This reconstructs the arrows in the sketch, which lead from a row and a column to their product.

To handle this model, consider synthetic control and synthetic differences-in-differences. Suppose $W_{it} = 1$ exactly when $i > n_0$ and $t > T_0$. The diagram partitions the outcome matrix at row n_0 and column T_0, with only the bottom-right block treated.

Fit the untreated units to the pretreatment trajectory of the treated group:

$$\sum_{i=1}^{n_0} \gamma_i Y_{it} \approx \alpha + \frac{1}{n - n_0} \sum_{i=n_0+1}^{n} Y_{it}, \qquad t = 1, \ldots, T_0.$$

The first group is never treated; the second is treated after T_0. The intercept α plays a role similar to a fixed effect. Choose weights by

$$\begin{aligned} (\widehat{\gamma}, \widehat{\alpha}) \in \operatorname*{arg\,min}_{\gamma', \alpha} \quad & \left\| \sum_{i=1}^{n_0} \gamma_i' Y_{i,1:T_0} - \frac{1}{n - n_0} \sum_{i=n_0+1}^{n} Y_{i,1:T_0} - \alpha \mathbf{1} \right\|_2^2, \\ \text{subject to} \quad & \sum_{i=1}^{n_0} \gamma_i' = 1, \qquad \gamma_i' \geq 0. \end{aligned} \tag{8.7}$$

My red annotation: the purpose of γ_i is to create parallel trends, so that the latent A_i terms can be balanced.

Finally, estimate the treatment coefficient in the weighted two-way model:

$$(\widehat{\tau}, \widehat{\alpha}, \widehat{\beta}) \in \operatorname*{arg\,min}_{\tau', \alpha', \beta'} \sum_{i,t} \gamma_i (Y_{it} - \alpha_i' - \beta_t' - W_{it}\tau')^2, \tag{8.8}$$

where $\gamma_i = 1/(n - n_0)$ for $i > n_0$. The closed-form treatment coefficient is

$$\widehat{\tau} = \frac{1}{n - n_0} \sum_{i=n_0+1}^{n} \left\{ \frac{1}{T - T_0} \sum_{t=T_0+1}^{T} Y_{it} - \frac{1}{T_0} \sum_{t=1}^{T_0} Y_{it} \right\} - \sum_{i=1}^{n_0} \gamma_i \left\{ \frac{1}{T - T_0} \sum_{t=T_0+1}^{T} Y_{it} - \frac{1}{T_0} \sum_{t=1}^{T_0} Y_{it} \right\}.$$

The red margin says this depends on T_0 and n. Under an appropriate large-panel asymptotic regime, $n_0, n_1, T \to \infty$, the estimator is consistent. A further handwritten qualification below this statement is clipped by the bottom edge of the scan; the visible fragments refer to the specification in (8.5) and asymptotic normal inference, but a complete sentence cannot be recovered.

Left-page postscript. Synthetic control is not the only method for interactive fixed effects. For example, apply k-means clustering to the rows of the outcome matrix Y, then fit a time-varying baseline model separately for each group. A short reference is written as "p66, literature" [page number/reference unclear].

Identification through exchangeability

Scan reference: Lecture1-10.pdf, page 43, with the extension on page 42

Recall $Y_{it} = Y_{it}(0) + W_{it}\tau$. A design-based assumption is

$$Y_{i\cdot}(0) \perp W_{i\cdot} \mid S_i, \qquad S_i = \sum_{t=1}^{T} W_{it}. \tag{8.10}$$

One implication: consider a linear estimator

$$\widehat{\tau} = \sum_{i,t} \gamma_{it} Y_{it}, \tag{8.11}$$

whose weight matrix depends only on W. Under (8.10), the untreated potential outcomes are exchangeable within a value of S_i, and

$$\mathbb{E}[\widehat{\tau} \mid W, \gamma] = \sum_{i,t} \gamma_{it} \mathbb{E}[Y_{it}(0) \mid S_i] + \tau \sum_{i,t} \gamma_{it} W_{it}.$$

Thus unbiasedness follows if

$$\sum_{i,t} \gamma_{it} W_{it} = 1, \qquad \sum_{i:S_i=s} \gamma_{it} = 0 \quad \text{for every } t \text{ and } s \in \mathcal{S},$$

where $\mathcal{S}$ is the support of S_i. **My annotation:** within a stratum the conditional expectation is constant, so the zero sum of weights removes that term.

Choose γ by minimizing variance under these unbiasedness constraints:

$$\begin{aligned} \gamma \in \operatorname*{arg\,min}_{\gamma'} \quad & \sum_{i,t} (\gamma_{it}')^2, \\ \text{subject to} \quad & \sum_{i,t} \gamma_{it}' W_{it} = 1, \quad \sum_{i:S_i=s} \gamma_{it}' = 0 \quad \text{for all } s, t. \end{aligned} \tag{8.14}$$

Blue note: roughly, we need variation in W_{it} among units with the same S_i.

Left-page extension: two sets of identifying restrictions. Recall the two-way model

$$Y_{it} = \alpha_i + \beta_t + W_{it}\tau + \varepsilon_{it}, \qquad \mathbb{E}[\varepsilon \mid \alpha, \beta, W] = 0. \tag{8.2, recalled}$$

The estimator (8.11) is also unbiased under this model if

$$\sum_t \gamma_{it} = 0 \text{ for every } i, \qquad \sum_i \gamma_{it} = 0 \text{ for every } t, \qquad \sum_{i,t} \gamma_{it} W_{it} = 1.$$

Red note: the first two equalities cancel the fixed effects α_i and β_t, and the third normalizes the treatment coefficient.

Combine the restrictions to obtain a doubly robust estimator: minimize $\sum_{i,t} \gamma_{it}^2$ subject to

$$\sum_{i,t} \gamma_{it} W_{it} = 1, \qquad \sum_{i:S_i=s} \gamma_{it} = 0 \quad \text{for all } s, t,$$

$$\sum_t \gamma_{it} = 0 \quad \text{for all } i, \qquad \sum_i \gamma_{it} = 0 \quad \text{for all } t.$$

It is unbiased if either (8.2) or (8.10) holds. An abbreviated reference near the top of the supplement reads "p67, [2019]" [the citation is not fully identifiable from the scan].

Lecture 9: Instrumental variables regression

Scan reference: Lecture1-10.pdf, page 45, with the three IV conditions on page 44

Instrumental variables regression addresses endogenous treatment effects, where $Y_i(0)$ and W_i are dependent.

A structural model

Start with a constant-effect model:

$$Y_i(w) = Y_i(0) + w\tau, \qquad Y_i = Y_i(W_i). \tag{9.1}$$

Unconfoundedness ordinarily means $\{Y_i(w)\} \perp W_i$; in (9.1), this is $Y_i(0) \perp W_i$. Rewrite the model as

$$Y_i = \alpha + W_i\tau + \varepsilon_i, \tag{9.2}$$

where $\alpha = \mathbb{E}[Y_i(0)]$ and $\varepsilon_i = Y_i(0) - \mathbb{E}[Y_i(0)]$. Under unconfoundedness, $\mathbb{E}[\varepsilon_i \mid W_i] = 0$, so OLS of Y_i on W_i estimates τ.

With confounding, equation (9.2) still holds, but $\mathbb{E}[\varepsilon_i \mid W_i] \neq 0$. OLS then targets

$$\begin{aligned} \tau_{\text{OLS}} &= \frac{\text{Cov}(Y_i, W_i)}{\text{Var}(W_i)} \\ &= \frac{\text{Cov}(\tau W_i + \varepsilon_i, W_i)}{\text{Var}(W_i)} = \tau + \frac{\text{Cov}(\varepsilon_i, W_i)}{\text{Var}(W_i)} \neq \tau. \end{aligned}$$

The ratio is underlined as the OLS coefficient.

Identification with an instrument

Introduce Z_i:

$$Y_i = \alpha + W_i\tau + \varepsilon_i, \qquad \varepsilon_i \perp Z_i, \qquad W_i = Z_i\gamma + \eta_i. \tag{9.4}$$

Since Z_i is uncorrelated with ε_i,

$$\text{Cov}(Y_i, Z_i) = \text{Cov}(\tau W_i + \varepsilon_i, Z_i) = \tau\,\text{Cov}(W_i, Z_i).$$

Hence

$$\tau = \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(W_i, Z_i)}, \qquad \widehat{\tau} = \frac{\widehat{\text{Cov}}(Y_i, Z_i)}{\widehat{\text{Cov}}(W_i, Z_i)}.$$

My left-page checklist for an IV.

1. **Exogenous:** $\varepsilon_i \perp Z_i$.
2. **Relevant:** $\mathrm{Cov}(W_i, Z_i) \neq 0$.
3. **Exclusion restriction:** every effect of Z_i on Y_i must pass through W_i.

The sketch is $Z_i \to W_i \to Y_i$; additional routes from treatment to outcome are sketched, while the instrument's influence is required to go through treatment.

Optimal instruments

Scan reference: Lecture1-10.pdf, page 47, with extensions on page 46 and the estimating-equation reminder on page 48

Allow a high-dimensional instrument $Z_i \in \mathcal{Z}$:

$$Y_i = \tau W_i + \varepsilon_i, \qquad \varepsilon_i \perp Z_i, \qquad Y_i, W_i \in \mathbb{R}.$$

For a function $w : \mathcal{Z} \to \mathbb{R}$,

$$\tau = \frac{\mathrm{Cov}(Y_i, w(Z_i))}{\mathrm{Cov}(W_i, w(Z_i))}, \tag{9.8}$$

with nonzero denominator. Which transformation $w(\cdot)$ is optimal? The estimator is

$$\begin{aligned} \widehat{\tau}_w &= \frac{\widehat{\mathrm{Cov}}(Y_i, w(Z_i))}{\widehat{\mathrm{Cov}}(W_i, w(Z_i))} \\ &= \frac{n^{-1}\sum_i (Y_i - \overline{Y})(w(Z_i) - \overline{w(Z)})}{n^{-1}\sum_i (W_i - \overline{W})(w(Z_i) - \overline{w(Z)})}, \end{aligned} \tag{9.9}$$

where $\overline{Y} = n^{-1}\sum_i Y_i$, with analogous definitions for the other averages. It solves the estimating equation

$$\frac{1}{n}\sum_i (w(Z_i) - \overline{w(Z)})\{Y_i - \overline{Y} - \widehat{\tau}_w(W_i - \overline{W})\} = 0. \tag{9.10}$$

The source writes

$$n(\widehat{\tau}_w - \tau) \Rightarrow \mathcal{N}(0, V_w), \qquad V_w = \frac{\mathrm{Var}(\varepsilon_i)\,\mathrm{Var}(w(Z_i))}{\mathrm{Cov}(W_i, w(Z_i))^2}. \tag{9.11, source scaling}$$

Translator's note: the multiplier is handwritten as n, but the estimating-equation reminder on page 48 uses $\sqrt{n}$. The usual nondegenerate normal scaling for this expression is $\sqrt{n}$; the mismatch is recorded explicitly.

Because $Z_i \perp \varepsilon_i$ and $\mathbb{E}[\varepsilon_i] = 0$,

$$\mathrm{Var}\{\varepsilon_i(w(Z_i) - \mathbb{E}[w(Z_i)])\} = \mathrm{Var}(\varepsilon_i)\,\mathrm{Var}(w(Z_i)).$$

The optimization written in the notebook is

$$w^*(\cdot) \in \arg\min_{w'} \frac{\mathrm{Cov}(W_i, w'(Z_i))^2}{\mathrm{Var}(w'(Z_i))}.$$

Its stated solution is

$$w^*(z) \propto \mathbb{E}[W_i \mid Z_i = z].$$

Red interpretation: $w^*(z)$ is the best prediction of W_i from Z_i. *Translator's note:* minimizing V_w requires *maximizing*, not minimizing, the displayed covariance-squared/variance ratio. The source's "arg min" and its intended optimal prediction are both retained, with this discrepancy made explicit.

Left-page extension: conditioning on covariates. The condition can be extended to $\varepsilon_i \perp Z_i \mid X_i$, giving

$$\tau(x) = \frac{\mathrm{Cov}(Y_i, w(Z_i) \mid X_i = x)}{\mathrm{Cov}(W_i, w(Z_i) \mid X_i = x)}.$$

The annotation mentions a doubly robust estimator of the average effect $\tau = \mathbb{E}[\tau(X)]$, and a random-forest estimator of $\tau(\cdot)$.

Another reminder on the same page: for independent X_1, X_2,

$$\begin{aligned}\mathrm{Var}(X_1 X_2) = {} & \mathrm{Var}(X_1)\,\mathrm{Var}(X_2) \\ & + \mathbb{E}[X_2]^2\,\mathrm{Var}(X_1) + \mathbb{E}[X_1]^2\,\mathrm{Var}(X_2).\end{aligned}$$

Starred supplement: estimating equations and the delta method. If functions ψ_i satisfy $\mathbb{E}[\psi_i(\theta)] = 0$, and $\widehat{\theta}$ is estimated by

$$\frac{1}{n}\sum_{i=1}^{n} \psi_i(\widehat{\theta}) = 0,$$

then, in the regular setting described by the note's nonparametric delta-method reminder,

$$\sqrt{n}(\widehat{\theta} - \theta) \Rightarrow \mathcal{N}(0, V), \qquad V = \frac{\mathrm{Var}(\psi_i(\theta))}{\mathbb{E}[\psi_i'(\theta)]^2}.$$

Cross-fitting and feasible estimation

Scan reference: Lecture1-10.pdf, page 49

The optimal instrument is $w^*(z) = \mathbb{E}[W_i \mid Z_i = z]$. The general procedure is:

1. Fit a nonparametric first-stage regression to estimate $\mathbb{E}[W_i \mid Z_i = z]$, obtaining $\widehat{w}(\cdot)$.
2. Use $\widehat{w}(\cdot)$ as the instrument in (9.9), obtaining $\widehat{\tau}_w$.

However, with many instruments this can create overfitting bias. If $\widehat{w}(Z_i)$ is fit using the entire data set, it need not be independent of ε_i: the fitted instrument can inherit the dependence of W_i on ε_i.

My blue example in the margin. Suppose Z_i is noise. The note writes

$$W_i = \alpha + \beta Z_i + \eta_i, \qquad Y_i = \gamma + \delta\widehat{W_i} + \varepsilon_i = \gamma + \delta(\widehat{\alpha} + \widehat{\beta} Z_i) + \varepsilon_i,$$

marks the fitted term as biased, and draws an arrow comparing it to $Y_i = \gamma + \delta W_i + \varepsilon_i$. The point of the annotation is that using an overfit first stage does not remove the problematic dependence. *Translator's note:* the margin labels the two expressions "equivalent," but does not supply conditions for that equivalence; it is retained as the author's warning/example rather than as a proved identity.

Use K-fold cross-fitting instead. With $k(i)$ the fold containing observation i,

$$\widehat{\tau} = \frac{\widehat{\mathrm{Cov}}(Y_i, \widehat{w}^{(-k(i))}(Z_i))}{\widehat{\mathrm{Cov}}(W_i, \widehat{w}^{(-k(i))}(Z_i))}. \tag{9.13}$$

The corresponding fitted instrument is independent of the observation's error under the stated sampling setup. In particular, if $\widehat{w}^{(-k(i))}(z)$ is mean-square consistent for $\mathbb{E}[W_i \mid Z_i = z]$, then (9.13) is asymptotically equivalent to the oracle version of (9.9). The red reminder writes this consistency as

$$\lim_{n\to\infty} \mathbb{E}[(\widehat{w}_n - \mathbb{E}[W \mid Z])^2] = 0,$$

where n is the sample size.

Nonparametric instrumental variables regression

Scan reference: Lecture1-10.pdf, page 50

The previous model assumed $Y_i = W_i\tau + \varepsilon_i$. Generalize it to

$$Y_i = g(W_i) + \varepsilon_i, \qquad Z_i \perp \varepsilon_i, \qquad Y_i, W_i \in \mathbb{R}, \quad Z_i \in \mathcal{Z},$$

where g is a general smooth function. **Blue warning:** we cannot simply run a nonparametric regression of Y_i on W_i, because

$$g(w) \neq \mathbb{E}[Y_i \mid W_i = w].$$

Translator's note: the introductory parenthesis in the scan says "Z_i acts linearly on Y_i," although the displayed earlier model is linear in W_i. The formula is the basis of the translation, and that wording discrepancy is recorded here.

Since $Z_i \perp \varepsilon_i$ and $\mathbb{E}[\varepsilon_i] = 0$,

$$\begin{aligned}\mathbb{E}[Y_i \mid Z_i = z] &= \mathbb{E}[g(W_i) + \varepsilon_i \mid Z_i = z] \\ &= \mathbb{E}[g(W_i) \mid Z_i = z] = \int_{\mathbb{R}} g(w) f(w \mid z)\, dw. \end{aligned} \tag{9.15}$$

Here $f(w \mid z)$ is the conditional density of W_i given $Z_i = z$.

A learning procedure:

1. Estimate $\widehat{f}(w \mid z)$ using cross-fitting.
2. Choose a suitable function class $\mathcal{G}$ and minimize the loss to estimate g:

$$\widehat{g} \in \operatorname*{arg\,min}_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^{n} \left\{ Y_i - \int_{\mathbb{R}} g(w) \widehat{f}^{(-k(i))}(w \mid Z_i)\, dw \right\}^2. \tag{9.16}$$

To solve this in practice, approximate g with a basis expansion,

$$g_K(w) = \sum_{k=1}^{K} \beta_k \psi_k(w),$$

where the ψ_k are predetermined basis functions. As K grows, g_K better approximates g. Then

$$\widehat{\beta} \in \operatorname*{arg\,min}_{\beta} \frac{1}{n} \sum_{i=1}^{n} \{ Y_i - \widehat{m}^{(-k(i))}(Z_i)^{\mathsf{T}} \beta \}^2, \tag{9.17}$$

with components

$$\widehat{m}_k^{(-k(i))}(Z_i) = \int_{\mathbb{R}} \psi_k(w) \widehat{f}^{(-k(i))}(w \mid Z_i)\, dw.$$

Bottom-margin interpretation: consistency for $g(\cdot)$ needs conditions; (9.15) is complicated, while (9.17) turns the problem into regression by minimizing the loss for each basis coefficient. The blue margin adds that using (9.17) comes at the cost of regularization. The remaining bottom-edge wording is clipped and cannot be fully read. A nearby note says "analogous to 9.13."

Lecture 10: Local average treatment effects

Scan reference: Lecture1-10.pdf, page 52, then the compliance table and derivation on page 51

Recall the Lecture 9 IV model:

$$Y_i = \alpha + W_i\tau + \varepsilon_i, \quad \varepsilon_i \perp Z_i, \qquad W_i = Z_i\gamma + \eta_i. \tag{10.1}$$

When Z_i is binary,

$$\begin{aligned}\tau &= \frac{\operatorname{Cov}(Y_i, Z_i)}{\operatorname{Cov}(W_i, Z_i)} \\ &= \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[W_i \mid Z_i = 1] - \mathbb{E}[W_i \mid Z_i = 0]}.\end{aligned} \tag{10.2}$$

Treatment effects under noncompliance

In an RCT with noncompliance, $Y_i \in \mathbb{R}$ is the ordinary outcome; $W_i \in \{0, 1\}$ is the treatment actually received and is not randomized; $Z_i \in \{0, 1\}$ is randomized treatment assignment.

The constant τ in (10.1) may not be appropriate, and treatment receipt may be related to the treatment effect. Define potential treatment receipt $W_i(0), W_i(1)$, with $W_i = W_i(Z_i)$ and $Y_i = Y_i(W_i(Z_i))$.

The IV validity assumptions are:

- **Exclusion restriction:** $Y_i(w, z) = Y_i(w)$. Assignment affects the outcome only through treatment receipt.
- **Exogeneity:** $\{Y_i(0), Y_i(1), W_i(0), W_i(1)\} \perp Z_i$. Assignment is randomized.
- **Relevance:** $\mathbb{E}[W_i(1) - W_i(0)] \neq 0$. Assignment affects treatment receipt.

There are four compliance types:

	$W_i(1) = 0$	$W_i(1) = 1$
$W_i(0) = 0$	Never-taker	Complier
$W_i(0) = 1$	Defier	Always-taker

Assume there are no defiers, that is, **monotonicity**:

$$W_i(1) \geq W_i(0) \quad \text{for every } i.$$

Then

$$\begin{aligned}&\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0] \\ &\quad = \mathbb{E}[Y_i(W_i(1)) \mid Z_i = 1] - \mathbb{E}[Y_i(W_i(0)) \mid Z_i = 0] \\ &\quad = \mathbb{E}[Y_i(W_i(1)) - Y_i(W_i(0))] \\ &\quad = \mathbb{E}[\mathbf{1}\{C_i = \text{complier}\}(Y_i(1) - Y_i(0))].\end{aligned}$$

If compliers exist, applying Bayes' rule gives

$$\begin{aligned}\tau_{\text{LATE}} &= \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[W_i \mid Z_i = 1] - \mathbb{E}[W_i \mid Z_i = 0]} \\ &= \mathbb{E}[Y_i(1) - Y_i(0) \mid C_i = \text{complier}].\end{aligned} \tag{10.3}$$

This is the complier average treatment effect.

My blue postscript. When the constant-effect model (10.1) fails and there are never-takers, always-takers, defiers, and so on, the individual contrast $Y_i(1) - Y_i(0)$ is not generally observable. In particular, a never-taker is never observed with $W_i = 1$. The marginal line writes an "IATE" expression as $\mathbb{E}[Y_i(1) - Y_i(0)]$ [the label and intended distinction between individual and average effects are not resolved in the scan].

Supply and demand

Scan reference: Lecture1-10.pdf, page 53

For each market $i = 1, \dots, n$, let $S_i(p, z)$ be its supply curve and $Q_i(p, z)$ its demand curve, where price $p \in \mathbb{R}$ and the instrument $z \in \{0, 1\}$. Assume supply is continuous and increasing in price, while demand is continuous and decreasing. The equilibrium price P_i is the unique solution to

$$S_i(P_i, Z_i) = Q_i(P_i, Z_i).$$

Thus Z_i is the IV, P_i is treatment, and $Q_i(P_i, Z_i)$ is the outcome Y_i. The red causal chain is $Z_i \to S_i \to P_i \to Q_i(P_i)$.

Conditions:

- Exclusion: $Q_i(p, z) = Q_i(p)$ for every p, z.
- Exogeneity: $\{Q_i(p), S_i(p, z)\} \perp Z_i$.
- Relevance: $\mathrm{Cov}(P_i, Z_i) \neq 0$.
- Monotonicity: $S_i(p, 1) \leq S_i(p, 0)$ for all relevant p [the source writes the equilibrium-price symbol in this condition].

If $Q_i(p)$ is differentiable, let $Q_i'(p)$ denote its derivative. The author's parenthetical qualification says that ordinary differentiability is not necessary: because $Q_i(\cdot)$ is monotone decreasing, a derivative in the distributional sense exists.

$$\begin{aligned}\tau_{\mathrm{LATE}} &= \frac{\mathbb{E}[Q_i \mid Z_i = 1] - \mathbb{E}[Q_i \mid Z_i = 0]}{\mathbb{E}[P_i \mid Z_i = 1] - \mathbb{E}[P_i \mid Z_i = 0]} \\ &= \frac{\displaystyle\int \mathbb{E}[Q_i'(p) \mid P_i(0) \leq p \leq P_i(1)] P[P_i(0) \leq p \leq P_i(1)]\, dp}{\displaystyle\int P[P_i(0) \leq p \leq P_i(1)]\, dp}.\end{aligned}$$

The denominator is annotated "total integral."

Derivation in the notebook.

$$\begin{aligned}&\mathbb{E}[Q_i \mid Z_i = 1] - \mathbb{E}[Q_i \mid Z_i = 0] \\ &\quad = \mathbb{E}[Q_i(P_i(1)) \mid Z_i = 1] - \mathbb{E}[Q_i(P_i(0)) \mid Z_i = 0] \\ &\quad = \mathbb{E}[Q_i(P_i(1)) - Q_i(P_i(0))] \\ &\quad = \mathbb{E}\left[\int_{P_i(0)}^{P_i(1)} Q_i'(p)\, dp\right] \\ &\quad = \int \mathbb{E}[Q_i'(p) \mid P_i(0) \leq p \leq P_i(1)] P[P_i(0) \leq p \leq P_i(1)]\, dp.\end{aligned}$$

Red note: interchange the integration order using Fubini's theorem, which is linked in the margin to exchanging the integral and expectation.

Threshold crossing and willingness to pay

Scan reference: Lecture1-10.pdf, page 55, with the diagram and decomposition on page 54

Suppose each subject has an unobserved endogenous variable U_i such that

$$W_i = \mathbf{1}\{U_i \geq c(Z_i)\},$$

where $c(z)$ is a cutoff function depending on the instrument. Without loss of generality, take $U_i \sim$ Unif$[0,1]$, giving

$$c(z) = 1 - P(W_i = 1 \mid Z_i = z).$$

The IV setup imposes:

1. Potential outcomes $Y_i(0), Y_i(1)$ with $Y_i = Y_i(W_i)$: the instrument has no direct route to the outcome.
2. $\{Y_i(0), Y_i(1), U_i\} \perp Z_i$: exogeneity.
3. $c(Z_i)$ has a nontrivial distribution: relevance.
4. $c(z)$ is càdlàg (right-continuous with left limits) and nondecreasing.

Translator's note: beside the fourth condition the source also says that W_i is nondecreasing in Z_i. With the written rule $W_i = \mathbf{1}\{U_i \geq c(Z_i)\}$ and increasing c, W_i instead decreases. The source's cutoff convention is retained, and this sign inconsistency is explicitly flagged.

Define the marginal treatment effect (MTE)

$$\tau(u) = \mathbb{E}[Y_i(1) - Y_i(0) \mid U_i = u]. \tag{10.6}$$

The IV estimand is a weighted average of $\tau(u)$. In the notes, $Z_i \sim \mathcal{N}(0,1)$, with density φ. Let $S \subset \mathbb{R}$ be the discontinuity set of c, and

$$c_-(z) = \lim_{a \uparrow z} c(a).$$

The displayed representation is

$$\tau_{\text{LATE}} = \frac{\displaystyle\sum_{z \in S} \left\{ \int_{c_-(z)}^{c(z)} \tau(u)\, du \right\} \varphi(z) + \int_{\mathbb{R} \setminus S} \tau(c(z)) c'(z) \varphi(z)\, dz}{\displaystyle\sum_{z \in S} (c(z) - c_-(z)) \varphi(z) + \int_{\mathbb{R} \setminus S} c'(z) \varphi(z)\, dz}.$$

This displays the IV estimand as a convex average of marginal treatment effects.

My left-page explanation of the weights. The drawing plots $c(z)$ against z with an open dot at the left limit and a filled dot at the value after a jump. The jump location belongs to S. At a jump, integrate $\tau(u)$ over the strip from $c_-(z)$ to $c(z)$ and multiply by $\varphi(z)$. On the continuous part, evaluate the treatment effect at $u = c(z)$ and weight it by $c'(z)\varphi(z)$. The arrows call $\varphi(z)$ a probability weight; the derivative represents the width contributed by the changing cutoff. The denominator adds the total weight. A second sketch shades the interval in u corresponding to the jump, making the du contribution explicit.

Bottom qualification in the source. The note says that a monotone $c(z)$ must have bounded variation, then writes

$$c(z) = c_0 + \int_{-\infty}^{z} c'(a)\, da,$$

where c' is a nonnegative Lebesgue-integrable function. *Translator's note:* that integral-only representation describes an absolutely continuous part. It does not by itself include the jumps explicitly allowed above (nor a possible singular continuous part). The original statement is retained as a qualification requiring care, rather than silently deleting the jump terms.

Single jumps, multiple jumps, and continuous cutoffs

Scan reference: Lecture1-10.pdf, page 57, with the derivative supplement on page 56

Example 1: one jump. For a cutoff otherwise constant with a single jump,

$$c(z) = c_0 + \delta_1 \mathbf{1}\{z \geq z_1\},$$

the source lists

$$\begin{array}{ll} \text{Never-taker:} & U_i < c_0, \\ \text{Complier:} & c_0 \leq U_i < c_0 + \delta_1, \\ \text{Always-taker:} & U_i \geq c_0 + \delta_1. \end{array}$$

Translator's note: these are the notebook's labels. With its increasing-cutoff convention and $W_i = \mathbf{1}\{U_i \geq c(Z_i)\}$, the middle group switches from treated to untreated as z rises; it is therefore opposite to the earlier $W_i(1) \geq W_i(0)$ complier direction. This is the same convention inconsistency noted above.

For K jumps,

$$c(z) = c_0 + \sum_{k=1}^{K} \delta_k \mathbf{1}\{z \geq z_k\},$$

and the note gives

$$\tau_{\text{LATE}} = \frac{\displaystyle\sum_{k=1}^{K} \mathbb{E}[\tau(U_i) \mid c_-(z_k) \leq U_i < c(z_k)](c(z_k) - c_-(z_k))\varphi(z_k)}{\displaystyle\sum_{k=1}^{K} (c(z_k) - c_-(z_k))\varphi(z_k)}.$$

It is a convex combination of the average effects in the switching strata. The weights depend on the size of each stratum in U space and on $\varphi(z_k)$. *Translator's note:* the numerator's first difference appears as $z_k - c_-(z_k)$ in the handwriting, while the denominator and the immediately preceding general expression use $c(z_k) - c_-(z_k)$. The consistent cutoff difference is displayed here, with the source discrepancy recorded.

Example 2: a continuous cutoff. If $c(z)$ has no jumps, the displayed formula becomes

$$\tau_{\text{LATE}} = \frac{\displaystyle\int_{\mathbb{R}} \tau(c(z))c'(z)\varphi(z)\,dz}{\displaystyle\int_{\mathbb{R}} c'(z)\varphi(z)\,dz}.$$

The covariance derivation is

$$\begin{aligned} \operatorname{Cov}(Y_i, Z_i) &= \operatorname{Cov}(Y_i(0) + (Y_i(1) - Y_i(0))W_i, Z_i) \\ &= \operatorname{Cov}((Y_i(1) - Y_i(0))W_i, Z_i) \\ &= \operatorname{Cov}((Y_i(1) - Y_i(0))\mathbf{1}\{U_i \geq c(Z_i)\}, Z_i) \\ &= \operatorname{Cov}(\tau(U_i)\mathbf{1}\{U_i \geq c(Z_i)\}, Z_i). \end{aligned}$$

Define

$$H(z) = \mathbb{E}[\tau(U_i)\mathbf{1}\{U_i \geq c(z)\}].$$

For standard Gaussian Z_i, the note uses

$$\operatorname{Cov}(H(Z_i), Z_i) = \mathbb{E}[H'(Z_i)].$$

Left-page question: distributional derivatives. The starred supplement asks: "Does $H'(z)$ denote the distributional derivative of $H(\cdot)$?" It writes, in a mixed ordinary/distributional notation,

$$-H'(z) = \begin{cases} \left\{ \displaystyle\int_{c_-(z)}^{c(z)} \tau(u)\, du \right\} \delta_z, & z \in S, \\ \tau(c(z))c'(z), & \text{otherwise.} \end{cases}$$

Here δ_z is a Dirac delta at z. The accompanying reminder gives the informal defining properties of a delta centered at zero: $\delta(x) = 0$ for $x \neq 0$ and $\int_{-\infty}^{\infty} \delta(x)\, dx = 1$. The question mark and the distributional nature of this supplement are retained.

Estimating the marginal treatment effect

Scan reference: Lecture1-10.pdf, page 58

At a smooth point of $c(z)$,

$$\tau(c(z)) = \frac{\dfrac{d}{dz}\mathbb{E}[Y_i \mid Z_i = z]}{\dfrac{d}{dz}P(W_i = 1 \mid Z_i = z)}. \tag{10.12}$$

Assume the derivatives are well behaved, the numerator and denominator can be estimated, and the denominator is nonzero.

Proof in the notebook.

$$\begin{aligned} \mathbb{E}[Y_i \mid c(Z_i) = c] &= \mathbb{E}[Y_i(0) + \mathbf{1}\{U_i \geq c\}(Y_i(1) - Y_i(0)) \mid c(Z_i) = c] \\ &= \mathbb{E}[Y_i(0) + \mathbf{1}\{U_i \geq c\}(Y_i(1) - Y_i(0))] \\ &= \mathbb{E}[Y_i(0)] + \int_c^1 \tau(u)\, du. \end{aligned}$$

The last step is marked "Fubini." Therefore

$$\tau(c) = -\frac{d}{dc}\mathbb{E}[Y_i \mid c(Z_i) = c].$$

By the chain rule,

$$\frac{d}{dz}\mathbb{E}[Y_i \mid Z_i = z] = \left.\frac{d}{dc}\mathbb{E}[Y_i \mid c(Z_i) = c]\right|_{c=c(z)} c'(z).$$

Since $U_i \sim \text{Unif}(0, 1)$ independently of Z_i,

$$c'(z) = -\frac{d}{dz}P(W_i = 1 \mid Z_i = z).$$

Combining these identities gives (10.12), as checked in the original.